14S 18650 BATTERY PACK ON PCB — BMS BREAKOUT / TRAINING BOARD
— Board: A4 landscape 297 x 210 mm, 4-layer
— 14 cells in series: 58.8V max / 50.4V nominal / 42V empty (below the 60V SELV limit)
— Cell sense: JST XH 15-pin (B-, C1..C14) through 60V PPTCs + per-cell 5V TVS
— Temperature: 7x 10k B3950 NTC, one between each pair of cells, JST XH 14-pin
— Pack output: M4 lugs PACK+ (after 5x20 10A fuse) and PACK-
— Per cell: red/black test loops + reverse-insertion LED

01_Cell_Module_Row1

File: 01_Cell_Module_Row1.kicad_sch

CELL MODULE A: CELLS 1 TO 7

02_Cell_Module_Row2

File: 02_Cell_Module_Row2.kicad_sch

CELL MODULE B: CELLS 8 TO 14

03_BMS_Interface_and_Thermal

File: 03_BMS_Interface_and_Thermal.kicad_sch

BMS INTERFACE: 15-WIRE CELL SENSE + 7x NTC

04_Power_Path_and_Protection

File: 04_Power_Path_and_Protection.kicad_sch

POWER PATH: FUSE, PACK TERMINALS, TVS, INDICATORS

05_Mechanical_and_Hardware

File: 05_Mechanical_and_Hardware.kicad_sch

MECHANICAL: MOUNTING HOLES & FIDUCIALS

All parts JLCPCB/LCSC sourced — LCSC field on every symbol
14S Li-ion 18650 (58.8V max / 50.4V nom) | IEC 62368-1 ES2 hazard >60V not reached
Battery Pack Breakout — 14S / A4

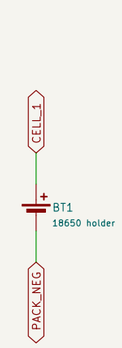
Sheet: /
File: BatteryPack_16S_A4_Breakout.kicad_sch

Title: 14S BATTERY PACK BREAKOUT — SYSTEM ARCHITECTURE

Size: A3	Date: 2026-10-12	**Rev: Rev 2.0 (14S)**
KiCad E.D.A. 10.0.6		Id: 1/6

CELL MODULE A: CELLS 1 TO 7
Each cell: 18650 holder, red (+) / black (–) test loops, and a zero-drain
reverse-insertion indicator (R -> LED -> 1N4148WS). It only conducts if the cell is
fitted backwards, so it draws nothing in normal use.

CELL 1







CELL 2







CELL 3







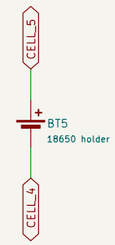
CELL 4







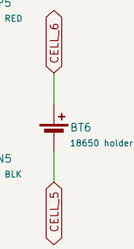
CELL 5

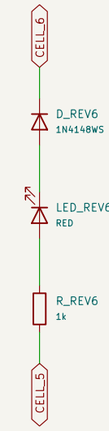






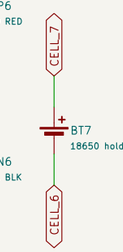
CELL 6

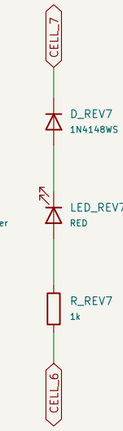


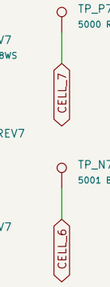




CELL 7







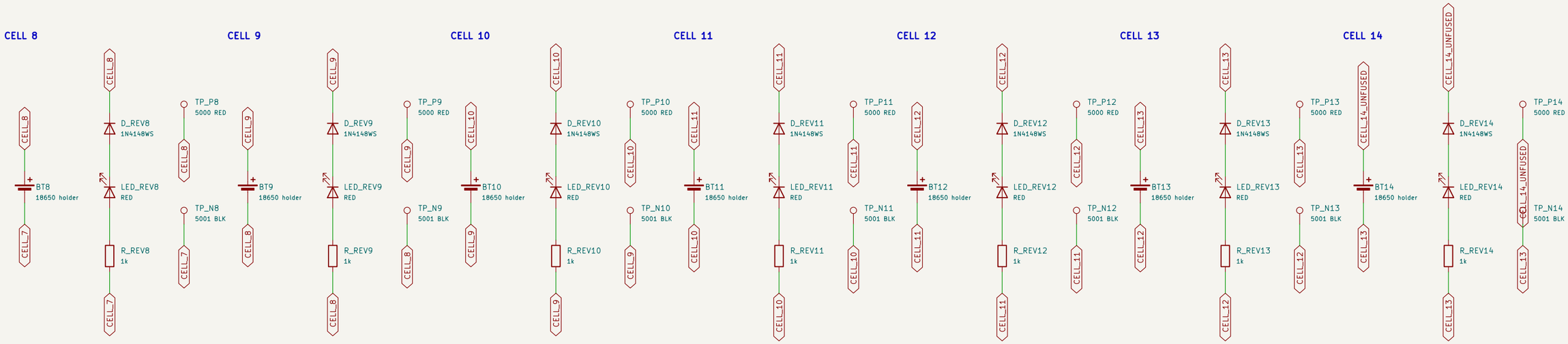
All parts JLCPCB/LCSC sourced – LCSC field on every symbol
14S Li-ion 18650 (58.8V max / 50.4V nom) | IEC 62368-1 ES2 hazard >60V not reached
Battery Pack Breakout – 14S / A4

Sheet: /01_Cell_Module_Row1/
File: 01_Cell_Module_Row1.kicad_sch

Title: CELL MODULE A: CELLS 1 TO 7

Size: A3	Date: 2026-10-12	**Rev: Rev 2.0 (14S)**
KiCad E.D.A. 10.0.6		Id: 2/6

CELL MODULE B: CELLS 8 TO 14
Each cell: 18650 holder, red (+) / black (−) test loops, and a zero-drain reverse-insertion indicator (R -> LED -> 1N4148WS). It only conducts if the cell is fitted backwards, so it draws nothing in normal use.



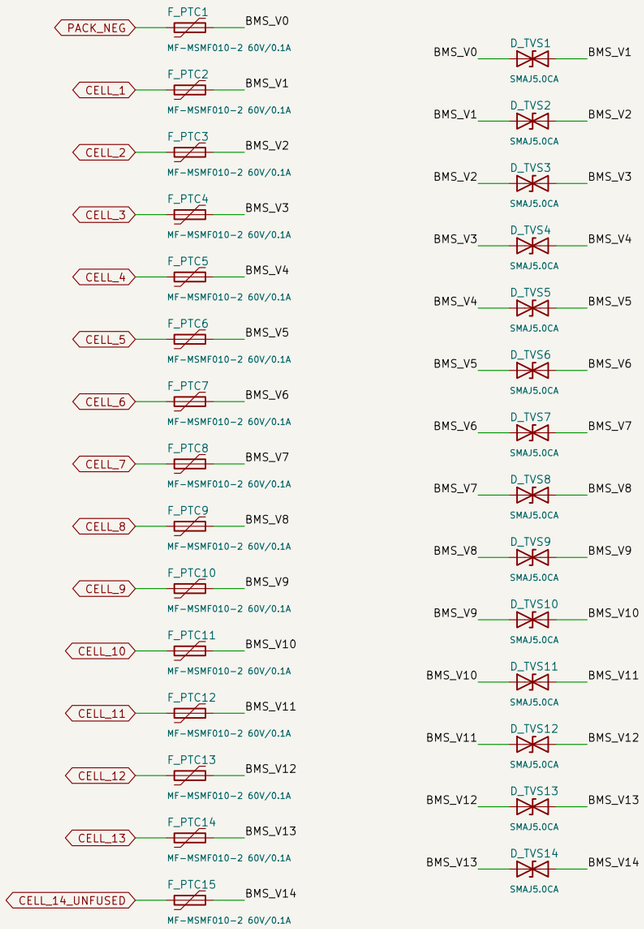
All parts JLCPCB/LCSC sourced — LCSC field on every symbol
14S Li-ion 18650 (58.8V max / 50.4V nom) | IEC 62368-1 ES2 hazard >60V not reached
Battery Pack Breakout — 14S / A4

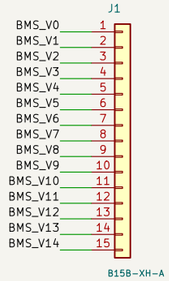
Sheet: /02_Cell_Module_Row2/
File: 02_Cell_Module_Row2.kicad_sch

Title: CELL MODULE B: CELLS 8 TO 14

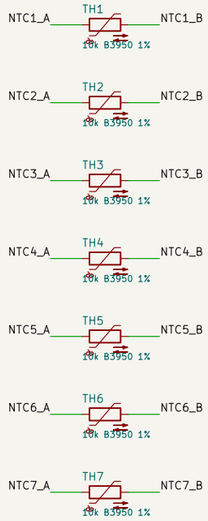
Size: A3	Date: 2026-10-12	**Rev: Rev 2.0 (14S)**
KiCad E.D.A. 10.0.6		Id: 3/6

BMS CELL-SENSE HARNESS (J1, JST XH B15B-XH-A, 15 pins)
Each tap: 60V-rated PPTC (MF-MSMF010-2) in series, 5V bidirectional TVS across every cell.
J1 pin 1 = B- (V0) ... pin 15 = C14+ (V14). Mating housing XHP-15 + SXH-001T-P0.6 crimps.





BMS TEMPERATURE HARNESS (J2, JST XH B14B-XH-A, 14 pins)
7x 10k B3950 NTC beads, one saddled between each cell pair, 2 isolated wires each.
J2 pin 2n-1 / 2n = NTCn. Mating housing XHP-14.



NTC1: saddle between cell 1 and cell 2

NTC2: saddle between cell 3 and cell 4

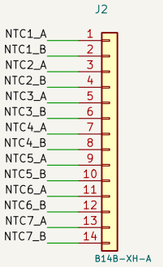
NTC3: saddle between cell 5 and cell 6

NTC4: saddle between cell 6 and cell 7

NTC5: saddle between cell 9 and cell 10

NTC6: saddle between cell 11 and cell 12

NTC7: saddle between cell 13 and cell 14



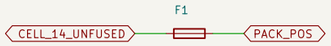
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14S Li-ion 18650 (58.8V max / 50.4V nom) | IEC 62368-1 ES2 hazard >60V not reached
Battery Pack Breakout — 14S / A4

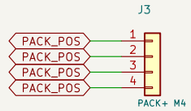
Sheet: /03_BMS_Interface_and_Thermal/
File: 03_BMS_Interface_and_Thermal.kicad_sch

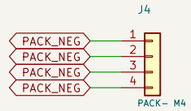
Title: BMS INTERFACE: 15-WIRE CELL SENSE + 7x NTC

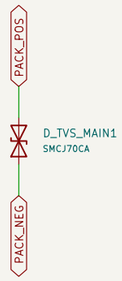
Size: A3	Date: 2026-10-12	**Rev: Rev 2.0 (14S)**
KiCad E.D.A. 10.0.6		Id: 4/6

POWER PATH: CELL_14 -> F1 (5x20mm, 10A) -> PACK+ (J3, M4 lug). PACK- = cell 1 negative (J4, M4 lug).
D_TVS_MAIN1 SMCJ70CA clamps pack transients (70V standoff > 58.8V max).





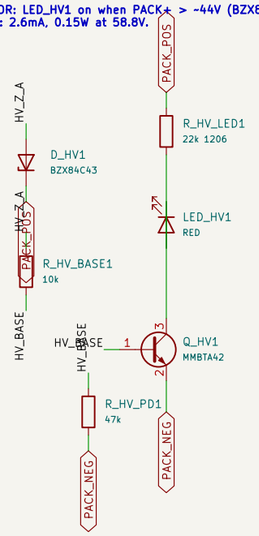




FUSE-BLOWN INDICATOR: lights amber when F1 is open and a load pulls PACK+ down.



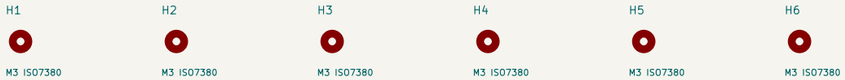
HV-PRESENT INDICATOR: LED_HV1 on when PACK+ > ~44V (BZX84C43 + Vbe).
R_HV_LED1 22k/1206: 2.6mA, 0.15W at 58.8V.



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14S Li-ion 18650 (58.8V max / 50.4V nom) | IEC 62368-1 ES2 hazard >60V not reached
Battery Pack Breakout — 14S / A4
Sheet: /04_Power_Path_and_Protection/
File: 04_Power_Path_and_Protection.kicad_sch
Title: POWER PATH: FUSE, PACK TERMINALS, TVS, INDICATORS

Size: A3	Date: 2026-10-12	**Rev: Rev 2.0 (14S)**
KiCad E.D.A. 10.0.6		Id: 5/6

MECHANICAL: 6x M3 holes (use nylon standoffs/screws — copper keep-out around each),
3x fiducials for JLCPCB SMT assembly.





All parts JLCPCB/LCSC sourced — LCSC field on every symbol
14S Li-ion 18650 (58.8V max / 50.4V nom) | IEC 62368-1 ES2 hazard >60V not reached
Battery Pack Breakout — 14S / A4

Sheet: /05_Mechanical_and_Hardware/
File: 05_Mechanical_and_Hardware.kicad_sch

Title: MECHANICAL: MOUNTING HOLES & FIDUCIALS

Size: A3	Date: 2026-10-12	**Rev: Rev 2.0 (14S)**
KiCad E.D.A. 10.0.6		Id: 6/6